

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 2: What Is Food Really Made Of? Introducing the Six Chemicals of Life

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	B	A	C	B	C	D

USE AFTER: PREDICT PHASE — Predict: written and pair predictions about the two Kisumu children (ugali vs githeri)

Q1. Two children in the same village in Kisumu eat about the same amount of food every day. Child A eats only ugali and develops a swollen belly and thin limbs (kwashiorkor). Child B eats githeri (maize and beans) and stays healthy. Which idea best explains the difference?

- A. Child B simply eats more food than Child A
- B. The two foods are made of different chemicals, and ugali has very little protein**
- C. Ugali has no nutrients at all
- D. Child A is ill only because he drinks too little water

Correct answer: B. Quantity is the same, so the difference must come from what the foods are made of. Ugali is mostly starch with very little protein, while githeri adds protein from beans.

USE AFTER: OBSERVE PHASE — Observe: Food Card Sort and the Six Chemicals of Life reference chart

Q2. Which list correctly names the six major chemicals of life?

- A. Carbohydrates, proteins, energy, nucleic acids, vitamins/minerals, water
- B. Carbohydrates, proteins, lipids, nucleic acids, vitamins/minerals, water**
- C. Starch, sugar, proteins, lipids, vitamins/minerals, water
- D. Carbohydrates, proteins, lipids, fibre, vitamins/minerals, water

Correct answer: B. The six major chemicals of life are carbohydrates, proteins, lipids, nucleic acids, vitamins/minerals and water. Fibre and energy are not chemical groups on this list, and starch and sugar are both carbohydrates.

USE AFTER: OBSERVE PHASE — Observe: second sort of food cards into the six chemical categories in the graphic organiser

Q3. During the food card sort, a learner places beans under carbohydrates because 'githeri is just a starch food'. Which response best corrects this idea?

- A. Beans are a rich source of protein, so they belong under proteins even though maize in githeri is mostly carbohydrate**
- B. Beans belong under vitamins because they are small and round
- C. Beans belong under water because they are cooked in water
- D. Beans belong under carbohydrates because all plant foods are mainly starch

Correct answer: A. Maize supplies mostly carbohydrate, but beans are a good source of protein. This is why githeri gives the body building chemicals that ugali alone lacks.

USE AFTER: EXPLAIN PHASE — Explain: Cause and Effect writing task on why the child eating ugali alone developed kwashiorkor

Q4. The child who eats only ugali gets plenty of energy from starch but still develops kwashiorkor. What is the best explanation?

- A. The child gets too much protein from ugali
- B. Starch is harmful and damages the body
- C. Carbohydrates supply energy, but ugali has almost no protein, so the body cannot build and repair its tissues**
- D. Energy is not needed by the body, so the starch is wasted

Correct answer: C. Carbohydrates give energy, while proteins build and repair muscles, enzymes and hormones. Without enough protein the body cannot build tissues, which causes the swollen belly and thin muscles of kwashiorkor.

USE AFTER: DQB PHASE — DQB: green 'What We Now Understand' sticky notes

Q5. A learner writes this on a green sticky note: 'I now understand that ugali has no nutrients at all.' Which statement is the correct improvement?

- A. Ugali is made only of water
- B. Ugali is rich in carbohydrate for energy, but it has too little protein to build and repair the body**
- C. Ugali is a poor food because it has no carbohydrate
- D. Ugali contains more protein than githeri

Correct answer: B. Ugali is about 75% carbohydrate and gives energy, so it does have nutrients. The problem is that it supplies very little protein.

USE AFTER: MODEL BUILDING PHASE — Model: first-draft molecular model of Ugali (Child A) and Githeri (Child B)

Q6. You draw a model with one circle for ugali and one for githeri, showing the six chemicals of life in each. Which feature would show the biggest chemical difference that explains the health of the two children?

- A. A larger water section in the ugali circle only
- B. A much larger carbohydrate section in the githeri circle than in the ugali circle
- C. A larger protein section in the githeri circle than in the ugali circle**
- D. No difference at all between the two circles

Correct answer: C. Githeri contains beans, so it has much more protein than ugali. This difference explains why one child developed kwashiorkor and the other stayed healthy.

USE AFTER: END OF LESSON — End of lesson: review of the graphic organiser (chemical, role, Kenyan food source)

Q7. Which pairing of a Kenyan food with the chemical of life it is especially rich in is correct?

- A. Sukuma wiki: lipids
- B. Omena: carbohydrates
- C. Ugali: proteins
- D. Avocado: lipids (fats and oils)**

Correct answer: D. Avocado is rich in lipids. Ugali is mainly carbohydrate, omena is a good protein source, and sukuma wiki is a source of vitamins and minerals.